

IEEE CEC 2026 · WCCI

TRACK 27: AUTOMATING CI & DYNAMIC ALGORITHM CONFIGURATION

Early-Stopping Thresholds for ES-HyperNEAT

A Data-Driven Approach from Fitness Dynamics

Case study: ES-HyperNEAT on the 10-class MNIST classification task

Part of a PhD on scaling adaptive-substrate neuroevolution · building on previous work TPE hyperparameter search (GECCO'24)

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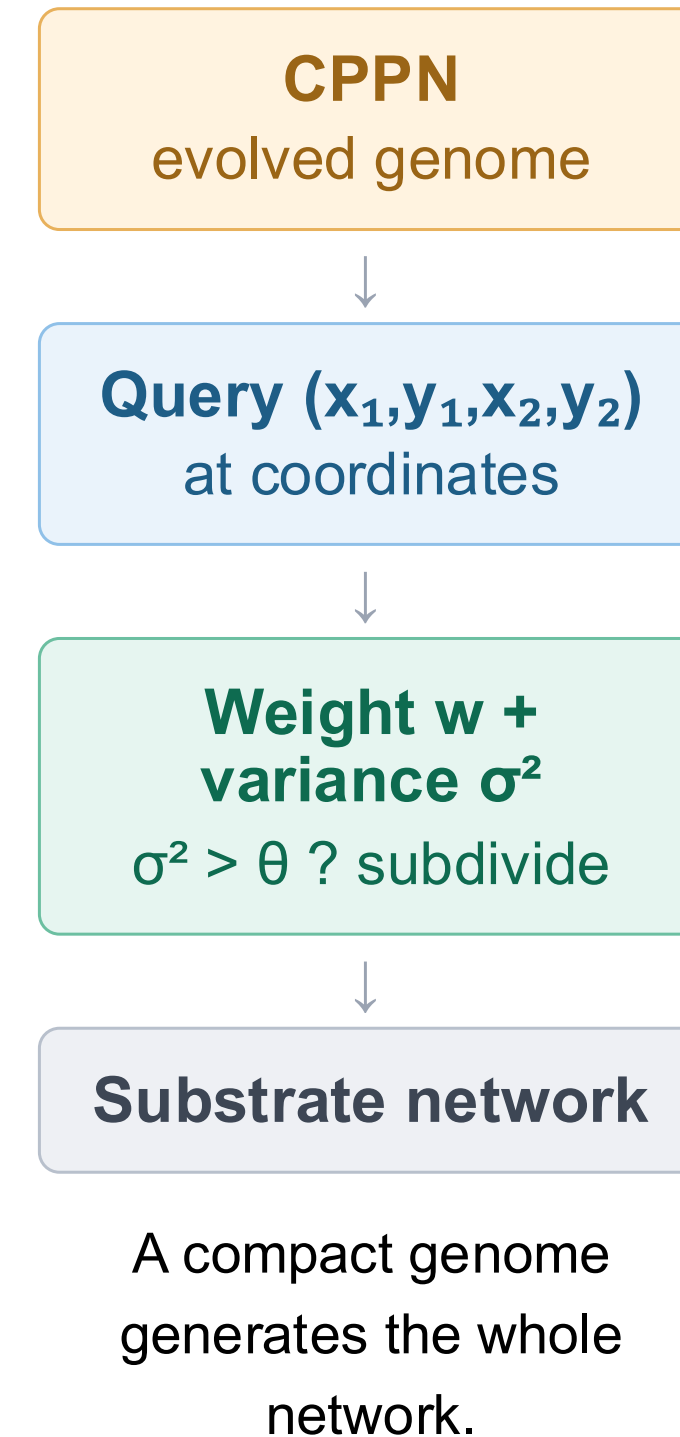
Most configurations never learn

ES-HyperNEAT (Risi & Stanley, 2012) evolves a neural network's **topology + weights** through an indirect **CPPN** encoding, with no backpropagation.

On our task, **10-class MNIST classification**, performance hinges on **16 interacting hyperparameters** (mutation rates, population size, substrate depth, variance / division thresholds). The value ranges **we scoped for this study** span over **3 billion configurations**, searched with a Bayesian optimizer (**TPE**).

The pain: ~80% of trials stagnate at the **0.1 random-guessing floor** (MNIST, 10 classes). They burn compute and inform the search nothing.

INDIRECT ENCODING



THE OPPORTUNITY, QUANTIFIED

Where the compute goes

93%

of trials sit at the 0.1 baseline at
generation 0

29%

never leave it across all 17
generations

100%

run to completion, we pay full price
for every one

The idea Detect dead-ends **early** and kill them, reallocating the budget to promising trials.

When can we decide?

How, without discarding the winners?

THREE QUESTIONS

RQ1 Can [early fitness dynamics](#) predict whether a trial will ultimately succeed or fail?

RQ2 How [early](#) in the run can that decision be made reliably?

RQ3 Does a [domain-specific](#) rule beat general-purpose schedules like [Hyperband](#)?

Our answer: [yes](#) · by [generation 3](#) · and [yes, at 64% lower cost](#).

The result is a **methodology**: it builds a **problem-specific rule** for ES-HyperNEAT on the MNIST benchmark under TPE search, but the approach applies to **other algorithms and tasks**.

Pruning as binary classification

Schedule-based pruners (SHA, Hyperband, BOHB, DEHB)

Drop a fixed fraction at preset resource budgets, **domain-agnostic**, ignoring *how this algorithm fails*.

Our move: read the algorithm's own dynamics

Given the **early median best fitness**, classify *success* vs. *failure*.

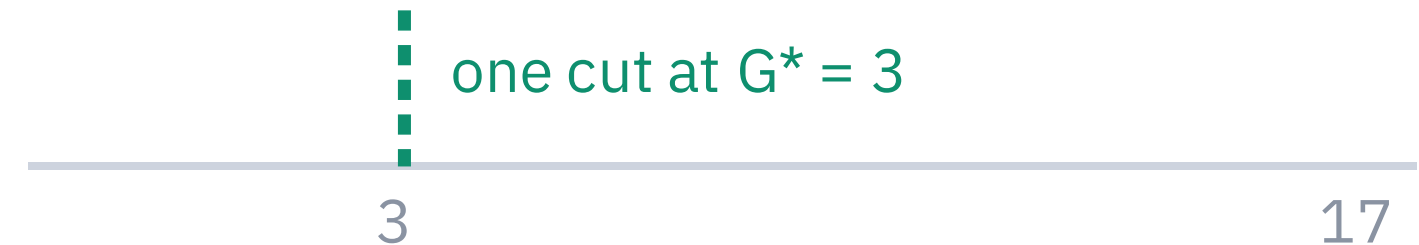
One feature → one threshold at one generation.

Hyperband: repeated rungs

drop a fixed fraction at preset budgets



Ours: one cut



A worked instance of the track's premise: a **configuration policy derived from algorithm behavior**, not imposed by a generic schedule.

METHOD: ONE METRIC, ONE RULE, TWO PHASES

METRIC

Cumulative median best fitness

outlier-resistant vs. mean / last

THE RULE

prune t at $G^* \Leftrightarrow \text{median best fitness}(t, G^*) \leq T^*$

PHASE 1 · DERIVATION

90 trials · 30 configs × 3 repeats

- run a **success-threshold sensitivity study**
- label with the chosen cutoff (top = “successful”, bottom = “unsuccessful”)
- choose **G^* , T^*** by maximizing F_1



PHASE 2 · VALIDATION

180 held-out trials

- counterfactual: **resume pruned trials** to completion
- measure precision / recall / F_1

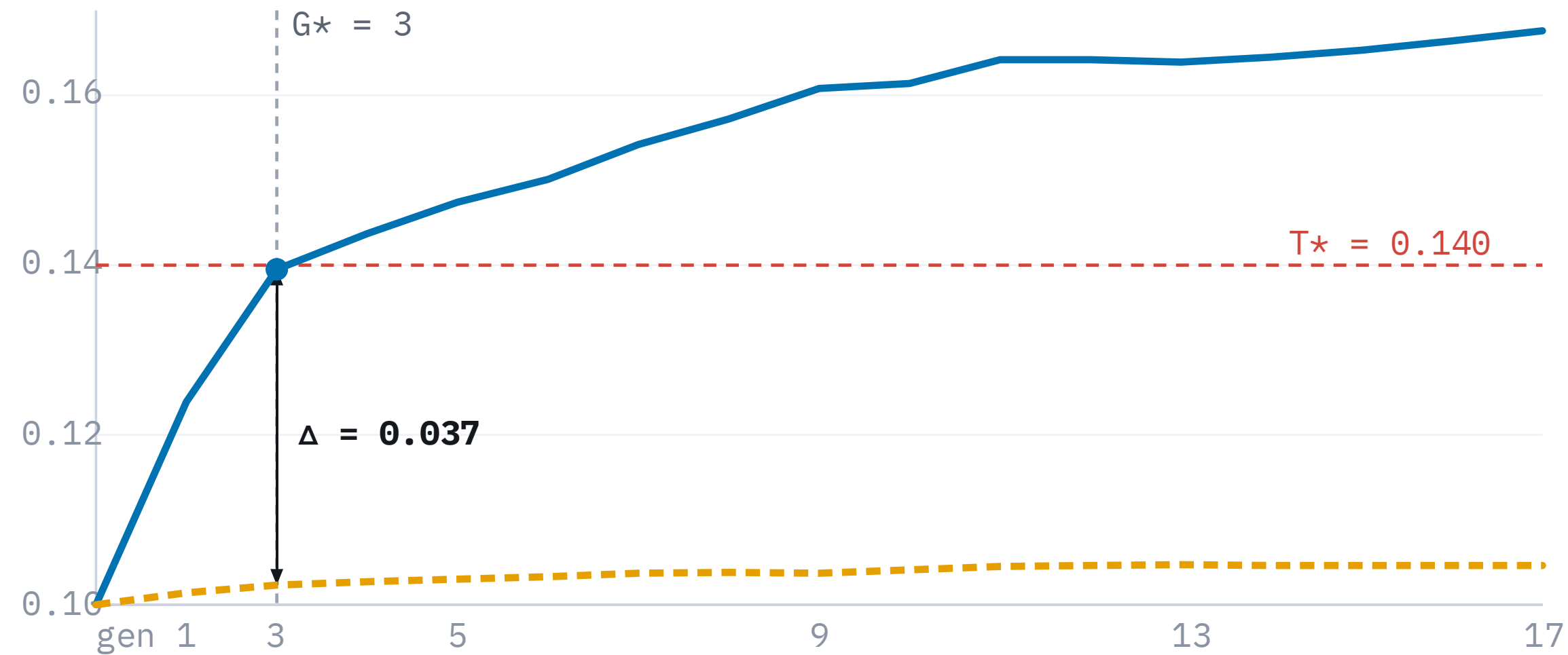
Setup: 10-class MNIST classification · 200 images/batch · 17 generations · CPU · PUREPLES + Optuna | cost reported in generations, not wall-clock

Success-threshold sensitivity

SUCCESS DEF.	RULE (G*, T*)	RECALL	PREC.	F ₁	SAVINGS
50th pct	(13, 0.145)	0.52	1.00	0.68	11.4%
75th pct	(4, 0.143)	0.86	0.97	0.91	38.2%
80th pct	(3, 0.140)	0.90	0.84	0.87	41.6%
90th pct	(3, 0.140)	0.94	0.70	0.80	41.6%

The **80th percentile** maximizes recall (don't discard winners) while delivering the largest savings. The 75th has higher F₁ but prunes later and keeps fewer winners.

Winners separate by generation 3



— **Successful** top 20%

— **Unsuccessful** bottom 80%

- Winners **jump immediately** and clear the baseline.
- Losers **stay flat** at ~ 0.10 forever.
- By **gen 3** the classes are cleanly split ($\Delta \approx 0.037$).

Mean of the median best fitness for trials labeled **Successful** (blue) and **Unsuccessful** (orange)
ES-HyperNEAT on MNIST, our scoped search space

VALIDATION: 180 HELD-OUT TRIALS

It generalizes to unseen trials

	PRED. KEEP	PRED. PRUNE
ACTUAL +	75 TRUE POS.	8 FALSE NEG.
ACTUAL -	14 FALSE POS.	83 TRUE NEG.

Pruned trials were resumed to reveal their true outcome.

0.872
F₁ score

0.904
Recall: keeps >90% of winners

0.843
Precision

0.757
MCC · acc. 0.878 · spec. 0.856

THE PAYOFF

41.6% cheaper, no quality loss

SAME 180 TRIALS · TOTAL GENERATIONS

No pruning 3,060



Our rule 1,786 -41.6%



Quality is preserved

TOST equivalence vs. unpruned baseline:

margin $\Delta = \pm 0.02$, observed diff -0.006 , **p < .001**.

⇒ statistically equivalent fitness at 42% lower cost.

ANOVA: our rule vs. unpruned, no significant difference (p = 0.069).

64% cheaper, higher mean fitness

TOTAL COST · GENERATIONS

Our rule 10,656

Hyperband 29,574

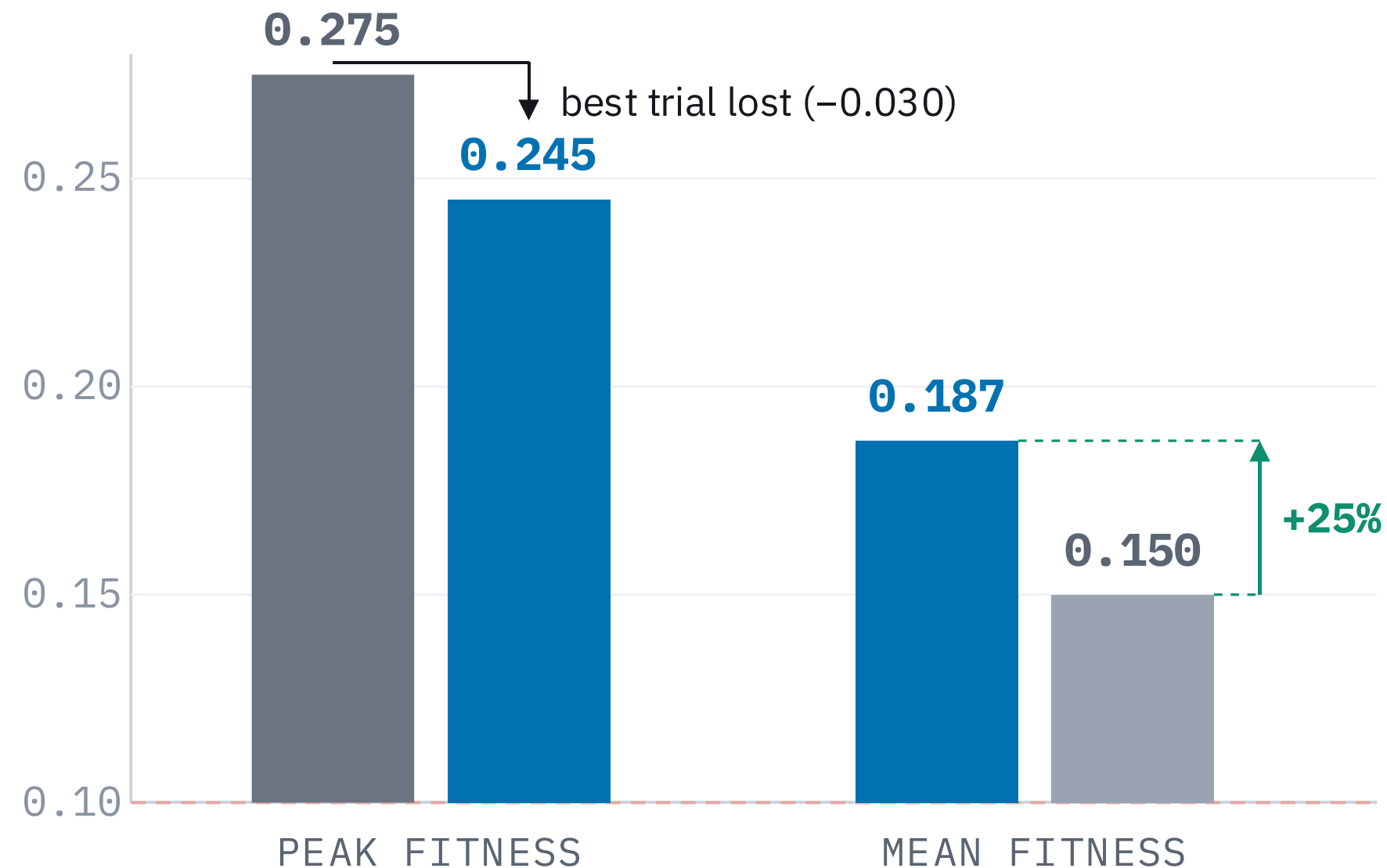
Unpruned 40,260

-64% vs. Hyperband

	OURS	HbP	BASE
Avg. top-5% fit.	0.206	0.195	0.241
Peak fitness	0.225	0.250	0.275

Honest trade-off: we win on average (Cohen’s d = 1.88); Hyperband’s broader search occasionally hits a higher single peak. Domain-specific wins when failure is diagnosable early.

Replayed on a converged search, the best trial is lost



GENERATIONS EVALUATED · FULL-TRIAL
EQUIVALENTS

No pruning 824 trials

Our rule 221 trials -73.2%

Still separates fit from unfit, but the threshold is **too strict** for an elite population: the single best trial (**0.275**) is a “slow starter” the rule discards, capping kept fitness at **0.245**.

Peak, best score across **all trials** (0.275) vs. among **kept** trials (0.245): the rule prunes the single best trial.

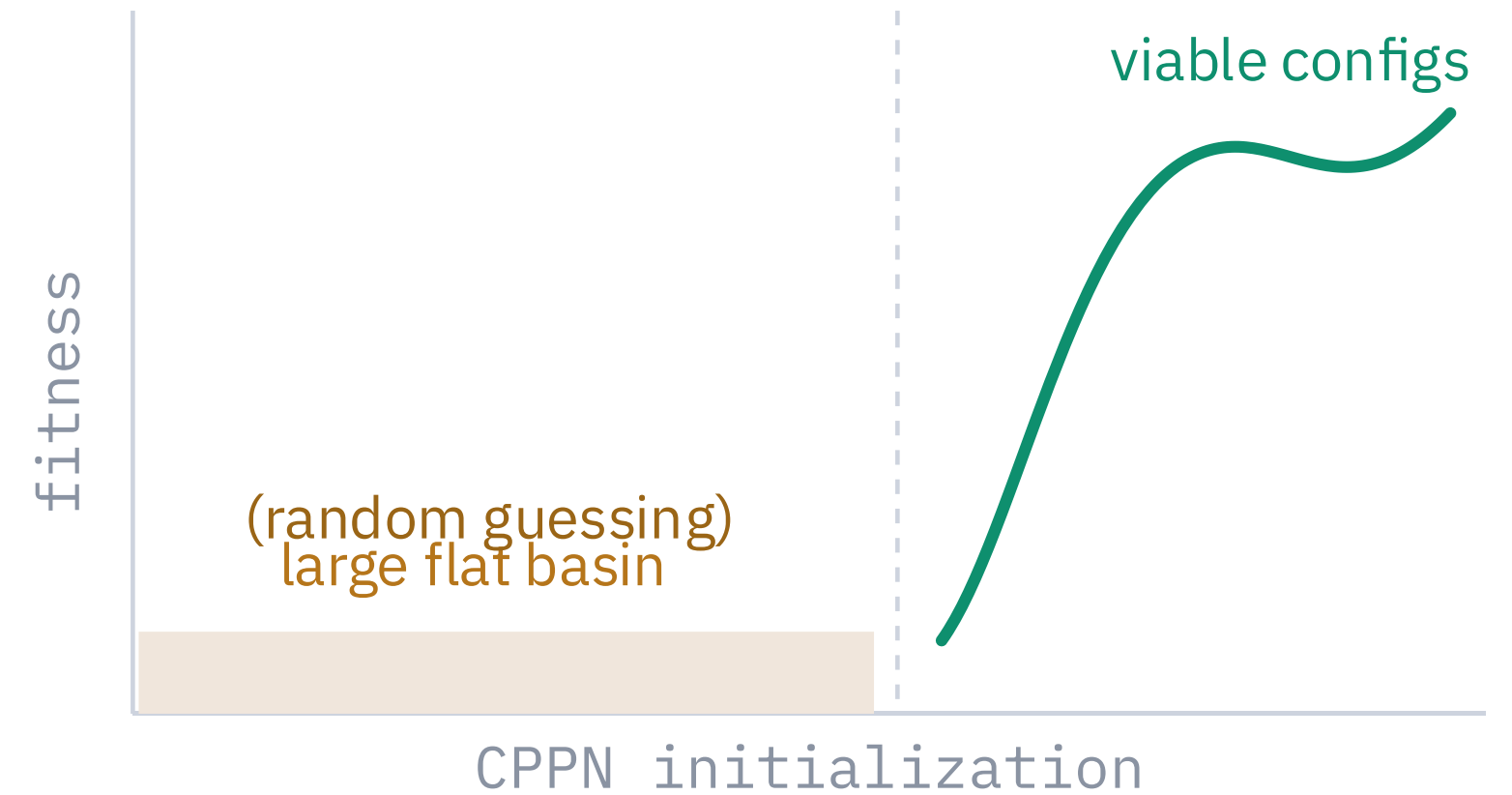
Mean, **kept** trials average higher (0.187) than **pruned** (0.150): the rule still separates good from bad.

All values from Table II, counterfactual simulation of our $G^*=3$ rule on 824 converged trials (Claret et al., 2024)

A “learn-or-don’t” fitness surface

The CPPN→substrate mapping produces a **binary** fitness landscape: a large flat basin at random guessing, and a separate region of viable configurations.

The outcome is largely **decided at initialization**. Small NEAT mutations rarely bridge a substrate with *no working signal path* to one that propagates input to output.



Two regions; initialization decides which one a trial lands in.

93%

start in the baseline basin

87%

stuck at gen 3 → stuck all 17 gens

<0.005

winners' change after the early jump

TAKEAWAYS

The methodology is the contribution

STEP 1

Frame pruning as **binary classification** on early fitness dynamics

STEP 2

Derive **one interpretable rule** : a decision generation and a threshold

STEP 3

Validate counterfactually, then **deploy to cut wasted compute**

Transfers to any evolutionary algorithm whose failures are diagnosable early; the rule's numbers are re-derived per task.

EXAMPLE VALUES · OUR CASE STUDY: ES-HYPERNEAT · 10-CLASS MNIST · TPE SEARCH

Rule: prune at gen 3 if median ≤ 0.140

$F_1 = 0.872$ on unseen trials

−41.6% cost, equivalent quality

−64% vs. Hyperband

→ **Next:** co-evolve the pruning rule alongside the search, so the threshold adapts as the population improves.

THANK YOU

Questions?

Early-Stopping Thresholds for ES-HyperNEAT

Romain Claret, Gygax, O'Neill, Cotofrei, Felber · University of Neuchâtel & University College Dublin

Derive stopping rules from fitness dynamics

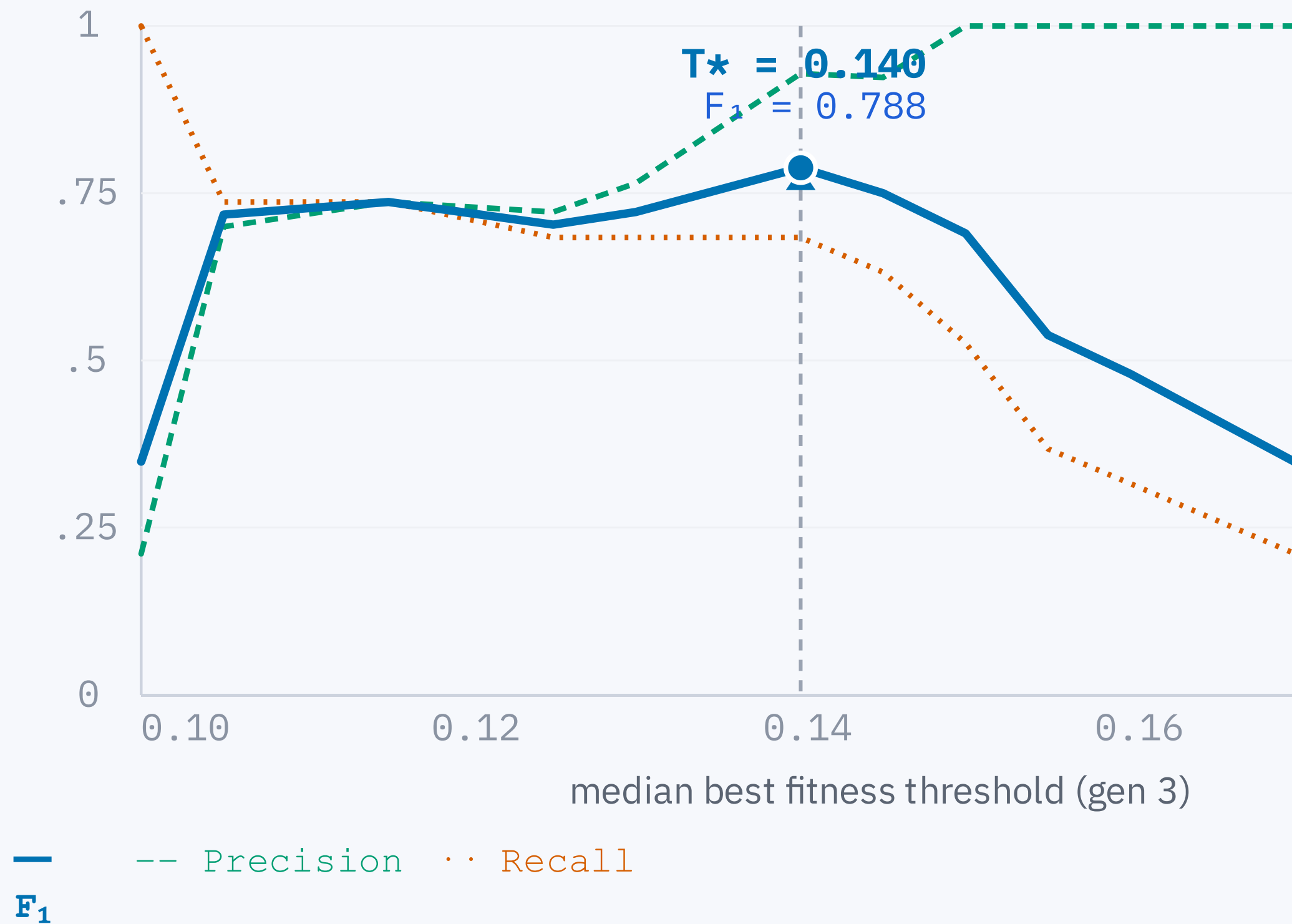
Keep the winners, cut the dead ends early

Cheaper search, equivalent quality

Outperforms fixed-schedule pruning

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F_1 peaks at $T^* = 0.140$



THE RULE

Kill any trial whose median best fitness at **generation 3** is

≤ 0.140

Stricter thresholds raise precision but cost recall, and 0.140 is the F_1 -optimal balance.

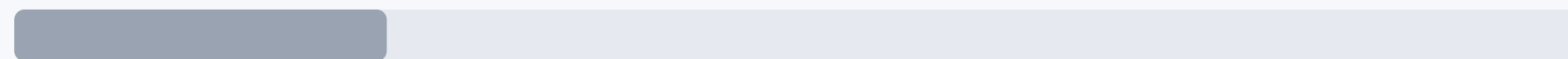
One generation, one number.

Problem-specific: these values hold for ES-HyperNEAT on MNIST in our scoped search space; the *methodology* transfers, the numbers don't.

Why $G^* = 3$ (robustness)

F_1 on validation set · axis 0.80–0.88

gen 2 0.819 [.754–.878]



gen 3 **0.872** [.817–.924]



gen 4 0.869 [.812–.917]



- **$G^*=3$ significantly beats $G^*=2$** : McNemar $p = 0.022$; paired-bootstrap F_1 diff [0.012, 0.098].
- **$G^*=3$ vs. $G^*=4$: no difference** ($p = 0.375$), so we take the **earlier** one.
- $G^*=3$ is the **earliest** generation reaching top performance.

Full statistical comparison

Omnibus: ANOVA $F(2, 3676) = 1090.7$, $p < .001$; Kruskal–Wallis $H(2) = 421.5$, $p < .001$, $\varepsilon^2 = 0.156$; parametric & non-parametric agree $\Rightarrow$ robust.

TUKEY HSD	MEAN DIFF	p	SIG.	COHEN'S d
Ours vs. Hyperband	0.052	<.001	yes	1.884
Unpruned vs. Hyperband	0.058	<.001	yes	1.555
Ours vs. Unpruned	−0.006	0.069	no	n/a

TOST equivalence (ours vs. unpruned): margin ± 0.02 , observed diff -0.0059 , $p < .001 \Rightarrow$ **statistically equivalent.**

The search space (extended)

HYPERPARAMETER GROUP	EXTENDED RANGE
Conn./node add & delete prob.	0.1–0.9 (step 0.1)
Population size	30–130 (step 10)
Initial depth / max depth	1–4 / 2–6
Variance / division thresholds	finer granularity
NEAT activation, init. connection, ...	base GECCO'24 space
Total	16 hyperparams, >3B configs

These value ranges define the search space **we scoped for this investigation** , extended from the GECCO'24 base space to widen exploration. **Most of this space produces non-learning networks** , which is exactly what makes early stopping valuable.

Why the *median* metric

Last-best fitness fails

The 80th percentile of last-best sits *at* the 0.1 baseline, with no signal to threshold on.

Median is outlier-resistant

A single lucky spike can't fake sustained learning; the median captures the trend.

Median vs. mean: no difference

Paired t-tests, $p > 0.05$, Cohen's $d < 0.2$, so we keep the median for robustness.

On a converged search space, the rule is too aggressive

Replay the rule on **824 elite, late-stage** trials from a converged TPE run: statistical separation still holds ($p < .001$), but “slow starters” have become common, and a threshold calibrated for random initial exploration discards them.

31.1%

recall on top-20% trials (was 90.4%)

0.245

peak: the best trial would be killed
(was 0.275)

73.2%

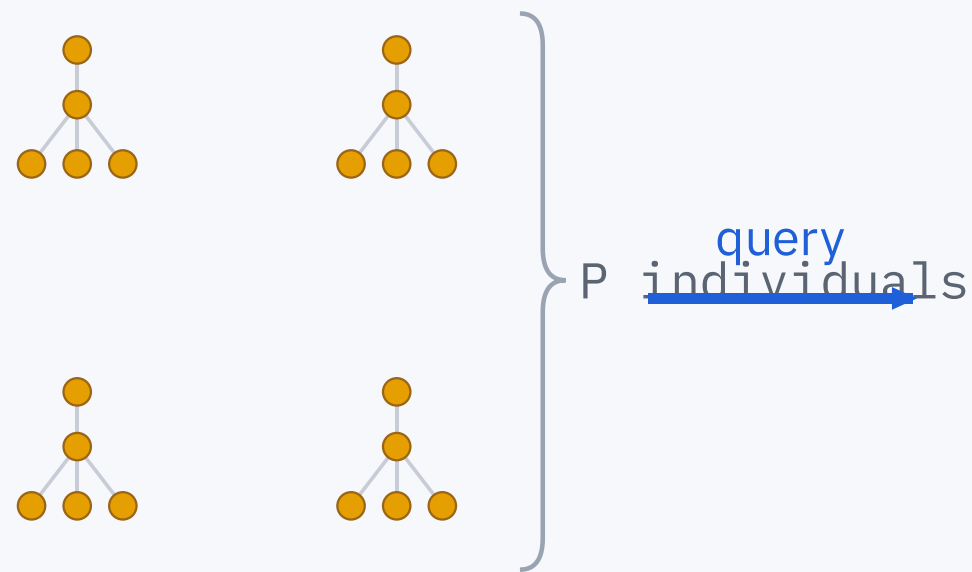
simulated savings: aggressive, but at
real cost

Takeaway: the rule is specialized for random initial exploration, motivating **adaptive thresholds** that relax as the search converges (BOHB / DEHB directions).

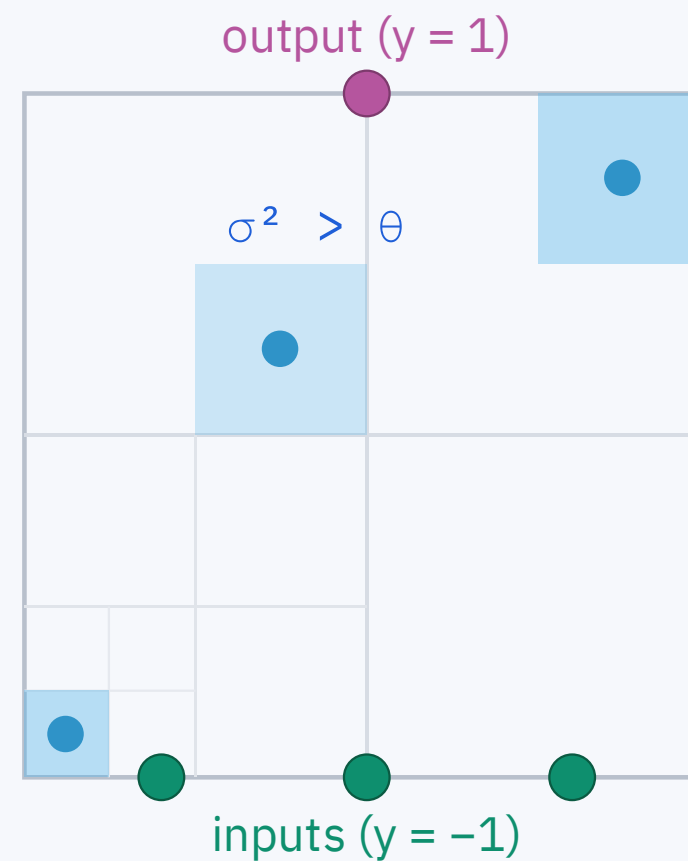
ES-HyperNEAT: from CPPN to substrate

CPPNs

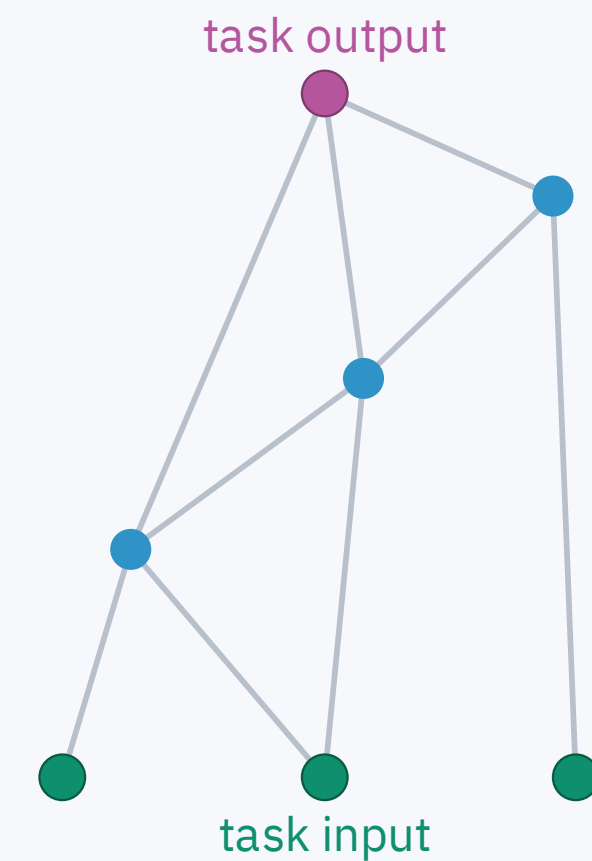
NEAT population · genotypes



substrate space



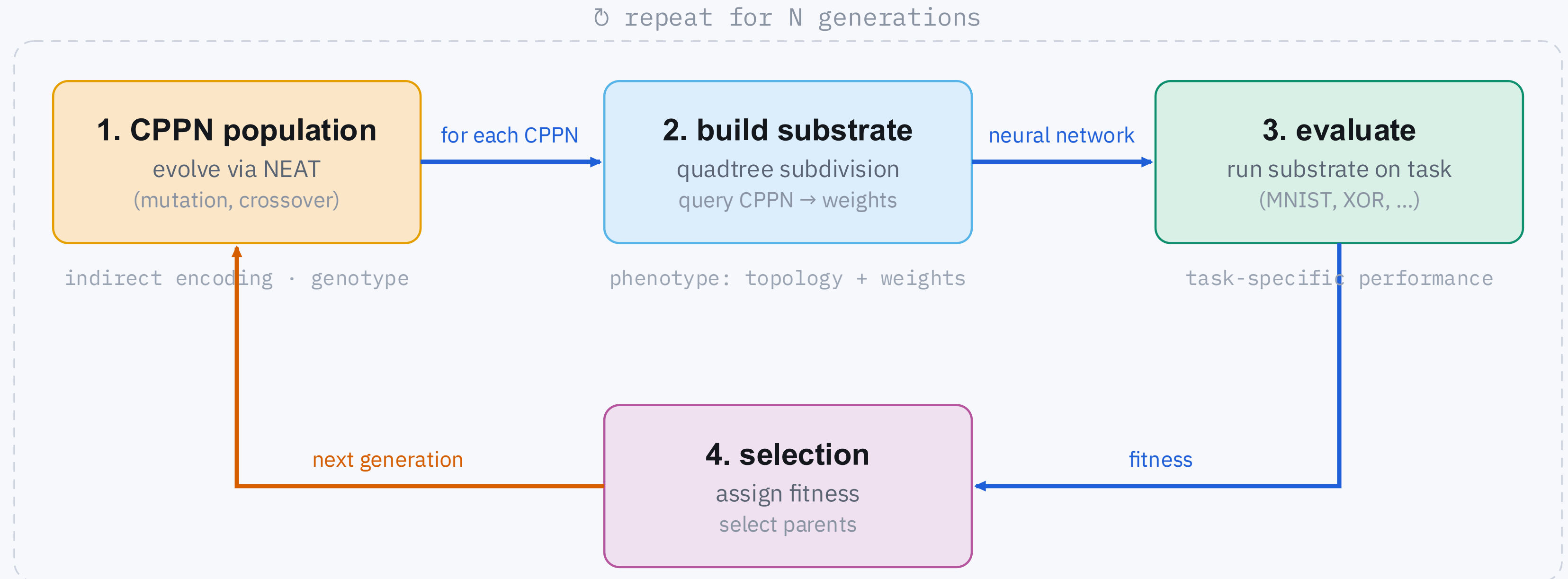
substrate network



● input ● hidden ● output ● CPPN node

Indirect encoding: a compact CPPN genome **generates** the whole network; the quadtree places hidden nodes where the CPPN's output varies most.

ES-HyperNEAT: the evolutionary loop



Key insight: each CPPN yields a *different* quadtree subdivision → a *different* substrate topology, the source of the irregular, stagnation-prone search space this paper tackles.